

Invited Mini Review

Decoding tau acetylation in Alzheimer's disease and tauopathies: from site-specific mechanisms to therapeutic horizons

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Tau acetylation has been recognized as a pivotal post-translational modification associated with the pathogenesis of Alzheimer's disease (AD) and other tauopathies. This review offers a detailed synthesis of the current understanding of site-specific tau acetylation, its regulatory enzymes, and its profound impacts on tau biology. Acetylation influences tau degradation, aggregation, propagation, and microtubule-binding properties in a residue-specific manner, often in conjunction with other modifications such as phosphorylation and ubiquitination. Furthermore, this review emphasizes emerging therapeutic strategies targeting tau acetylation, including small-molecule inhibitors of p300/CBP and monoclonal antibodies against acetylated tau epitopes. While several of these approaches are currently undergoing clinical trials, many acetylation sites are still inadequately characterized, emphasizing the need for additional mechanistic studies. An enhanced understanding of tau acetylation will be vital for devising targeted therapies to halt or reverse the progression of tau-mediated neurodegeneration. [BMB Reports 2025; 58(8): 325-339]

INTRODUCTION

Tau is a microtubule-associated protein (MAP) encoded by the *MAPT* gene located on chromosome 17. It is predominantly found in neurons, particularly within axons, though it also appears in lesser amounts in oligodendrocytes and astrocytes (1). Tau plays an essential role in the maintenance and stabilization of microtubule structures, which supports efficient axonal transport, promotes neurite outgrowth, and sustains

synaptic function (2, 3).

Tau was initially identified in 1985 as the main component of neurofibrillary tangles (NFTs), which are filamentous, intracellular aggregates made up of hyperphosphorylated tau proteins (4). As key pathological markers, NFTs are characteristic of primary tauopathies including frontotemporal lobar degeneration with tau pathology (FTLD-tau), progressive supranuclear palsy (PSP), and corticobasal degeneration (CBD). These tangles are also observed in secondary tauopathies such as Alzheimer's disease (AD) and chronic traumatic encephalopathy (CTE) (5, 6). In AD, NFTs are one of the two major pathological lesions, alongside amyloid plaques primarily composed of amyloid- β (A β) peptides. The amyloid cascade hypothesis suggests that the early accumulation of extracellular A β plaques initiates AD pathology (7). However, tau aggregation correlates more strongly with cognitive decline, highlighting tau's pivotal role in the progression of the disease (8). Additionally, the independent role of pathological tau in driving neurodegeneration is demonstrated by several tauopathies beyond AD that are often linked to mutations in the *MAPT* gene, occurring without amyloid deposits (8-10). Given the limited effectiveness of A β -targeted therapies (11), there is a growing interest in exploring alternative disease mechanisms, particularly those involving tau, as promising therapeutic targets (12).

Post-translational modifications (PTMs) of tau are crucial both for its physiological regulation and for its pathological dysregulation (6, 13). Various PTMs such as phosphorylation, acetylation, ubiquitylation, glycosylation, have been thoroughly explored for their roles in tau functionality and disease progression. Acetylation has become recognized as a primary modification, ranked second to phosphorylation in research prominence, largely because of its profound link with tauopathies (14), particularly AD. Tau acetylation can be detected in the early phases of AD (15-17) and intensifies as the disease progresses (18). Since first being identified in 2010 (15), tau acetylation has been acknowledged as a crucial PTM influencing tau degradation, aggregation, propagation, and neuronal toxicity. Notably, these effects vary depending on the acetylated lysine residues. Thus, understanding the functional consequences

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of site-specific acetylation is vital for pinpointing targeted therapeutic interventions.

In this review, we present a comprehensive overview of tau acetylation, emphasizing its site-specific roles in tau pathology and its potential as a therapeutic target. We also discuss current clinical strategies targeting tau acetylation, underscore existing challenges, and suggest prospective paths for the development of therapies based on acetylation.

THE STRUCTURE AND ISOFORMS OF TAU

Tau is predominantly a hydrophilic protein characterized by extensive intrinsically disordered regions (19–21). Like other MAP family proteins, its primary structure is organized into four major domains: the N-terminal projection domain, the proline-rich region (PRR), the microtubule-binding region (MTBR), and the C-terminal domain (Fig. 1). The N-terminal projection domain, which is intrinsically disordered, exhibits a net negative charge under physiological conditions (22). Adjacent to this, the PRR, more structured and positively charged, includes seven proline-X-X-proline motifs. These motifs include multiple phosphorylation sites (23) and serve as docking sites for signaling proteins with Src homology 3 (SH3) domains (24, 25). The MTBR, which comprises three or four repeats, is essential for tau's interaction with microtubules (26, 27). Notably, the VQIINK and VQIVYK motifs within this region are crucial for β -sheet formation, constituting the minimal sequences required for tau aggregation into paired helical filaments (PHFs) (28), which are fibrous polymers of tau distinguished by a twisted morphology. The C-terminal domain, similar to the N-terminus, is intrinsically disordered and features variably charged subregions that facilitate microtubule assembly (26, 29). Overall, the asymmetric charge distribution of tau underpins its interactions with microtubules and other binding partners, as well as its tendency to undergo internal folding and aggregation (30).

In the human central nervous system, alternative splicing of eight out of sixteen exons of the *MAPT* gene results in the production of six tau isoforms (31, 32). The N-terminal projection domain can be alternatively spliced to include 0 (0N), 1 (1N), or 2 (2N) inserts, while the MTBR can feature either three-repeat (3R) or four-repeat (4R) domains (26). Consequently, the six major tau isoforms are designated as 0N3R, 1N3R, 2N3R, 0N4R, 1N4R, and 2N4R (Fig. 1).

Tau isoforms display distinct expression patterns across various cell types, tissues, and subcellular compartments (26, 27). In the adult human brain, 3R and 4R tau isoforms are typically present at comparable levels (27), mirroring a developmental shift where 3R tau predominates during early stages, and 4R tau becomes more prevalent in adulthood (33). 4R tau demonstrates a greater capacity to enhance microtubule assembly compared to the 3R isoform (33). In numerous tauopathies, the balance between 3R and 4R isoforms is altered in a disease-specific manner. This alteration significantly impacts the aggrega-

tion properties of tau, resulting in neurofibrillary tangles that primarily consist of either 3R or 4R tau. These patterns underpin the classification of certain tauopathies as either 3R-specific or 4R-specific (34, 35). For example, PSP and CBD are associated with elevated levels of 4R tau, while Pick's disease is marked by an abundance of 3R tau (36). Conversely, AD, CTE, and primary age-related tauopathy (PART) display both isoforms in variable proportions that change throughout the course of the disease without a consistent pattern (12, 34, 35, 37).

TAU ACETYLATION

Tau acetylation was initially identified by Min *et al.* in both mouse models and human samples of AD (15). This discovery has since led to extensive research into the pathological significance of this PTM within the context of tau biology (14, 15, 38, 39). Tau acetylation has been observed across various tauopathies and is particularly evident at K280. It co-localizes with established tau pathology markers, with its presence becoming progressively pronounced as the disease advances to moderate and severe stages (14).

Among the various tauopathies, tau acetylation demonstrates a particularly strong correlation with AD. Site-specific acetylation of tau has been observed in the brains of AD patients even at early Braak stages and is notably abundant in NFTs (15–17). Additionally, tau acetylation levels are closely correlated with disease progression, increasing sharply during early and moderate Braak stages and becoming extensively associated with late-stage AD (18). Mass spectrometry analyses of wild-type and APP transgenic mice have identified multiple acetylated lysine residues in endogenous tau, further underscoring the importance of this modification in AD pathology (40).

Enzymes involved in tau acetylation

The enzymes responsible for tau acetylation in the brain are grouped into two categories: acetylating enzymes and deacetylating enzymes. Histone acetyltransferases (HATs), particularly p300 (15, 41–43) and CREB-binding protein (CBP) (44), play crucial roles in mediating tau acetylation (Fig. 1). Conversely, tau deacetylation is modulated by NAD⁺-dependent sirtuin 1 deacetylase (SIRT1) (15, 41) and histone deacetylase 6 (HDAC6) (42, 45, 46). While both SIRT1 and HDAC6 have been implicated in tau deacetylation, their mechanisms of action differ significantly. SIRT1 specifically targets acetylated lysines within the amino acid sequences 160–182 and 264–287 (15), which contributes to the prevention of tau accumulation *in vitro* (47, 48). In contrast, HDAC6 aggravates tau pathology by promoting its deacetylation (42, 46), leading to increased tau phosphorylation and subsequent aggregation *in vitro* (47, 48).

The role of acetyl-coenzyme A-induced auto-acetylation of tau *in vitro* remains contentious. Although studies have reported intrinsic enzymatic activity in tau, enabling auto-acetylation at the Lys280 residue through catalytic cysteine residues within the MTBR (49), these findings are challenged by other studies.

Subsequent investigations have not detected tau auto-acetylation in the absence of HATs, questioning the ability of tau to self-acetylate independently (42).

Dysregulation of tau acetylation via aberrant HATs and HDACs activities has been observed in various neurological diseases (16, 50). For instance, elevated p300/CBP HAT activity has been observed in the brains of patients with FTLD-tau and in the hippocampal CA1 region of AD brains (50, 51). Furthermore, in AD, the balance of acetylation is disturbed due to altered activities of HDAC6 and SIRT1. AD patients exhibit decreased SIRT1 expression (52) and increased HDAC6 levels (53), underscoring the contention that these enzymes play opposing roles in modulating tau function and contributing to tau pathology (13).

Acetylation sites of tau

Tau is acetylated at its lysine residues, particularly within the MTBR. The 2N4R tau isoform contains 44 lysine residues, indicating that approximately 10% of the protein is susceptible to acetylation (13). Analysis of postmortem brain tissue from AD patients has identified 19 distinct acetylated lysine residues, highlighting the significant prevalence of tau acetylation in the AD brain (Fig. 1) (18). The specific residues targeted for acetylation and the degree of their modification are influenced by a range of factors, including disease heterogeneity, stage of progression, tau isoform, and cellular context. These factors manifest as distinct acetylation patterns across different tauopathies and disease stages. For example, increased acetylation at lysine 274 has been reported in multiple tauopathies; however, its absence in argyrophilic grain disease distinguishes this condition from other tauopathies (39). These findings emphasize the importance of further research into the role of site-specific tau acetylation in the pathogenesis and classification of distinct tauopathy subtypes.

Interactions with other PTMs

PTMs of tau exhibit site-specific interactions that are influenced by the cellular environment, indicating that their effects are contingent not only upon the modified residue but also on surrounding biological conditions. Rather than being driven by a single modification, tau-related pathologies likely result from a complex interplay of structural and functional alterations that collectively enhance tau's pathogenic potential. Acetylation is particularly noteworthy, as many lysine residues targeted for this modification also serve as sites for other PTMs, such as ubiquitination, SUMOylation, methylation, and glycation (40, 54, 55). This convergence suggests potential competition among PTMs for the same lysine residues, wherein the presence of one modification may inhibit the addition of another, thereby contributing to a tightly regulated and dynamic system governing tau function (54).

Among these PTMs, ubiquitination is regarded as the most functionally significant competitor of acetylation. Acetylation inhibits ubiquitination at shared lysine residues, triggering a

cascade of downstream disruptions in tau homeostasis and pathogenicity, as elaborated in subsequent sections of this review (15, 16, 47). Notably, extensive overlap between acetylated and ubiquitinated residues is observed in both transgenic AD mouse models and post-mortem human AD brains. This convergence is particularly pronounced within the MTBR, where 7 out of 19 acetylated lysine sites also undergo ubiquitination (18, 40). Moreover, the interplay between tau ubiquitination and acetylation may contribute to the higher incidence of tauopathies, such as AD, observed in women (56). Ubiquitin specific peptidase 11 (USP11), an X-linked deubiquitinase that escapes X-chromosome inactivation and is thus more abundantly expressed in females, has been shown to remove ubiquitin moieties from tau (56). This deubiquitination unmasks lysine residues—including K174 and K281—rendering them susceptible to acetylation by p300/CBP. The resulting acetylation impairs proteasomal degradation of tau and promotes its pathological accumulation (56). Notably, genetic ablation of USP11 in female PS19 mice markedly reduced tau deposition, attenuated neuroinflammatory responses, and restored deficits in synaptic plasticity and cognitive performance. These findings implicate USP11—and the acetylation it facilitates—as a sex-specific driver of tau pathology (56).

While the relationship between tau acetylation and phosphorylation has been extensively studied, their interaction is complex and appears potentially dependent on specific residues and disease conditions. In AD brains, the distribution pattern of acetylated tau is similar to that of hyperphosphorylated tau (14). Though phosphorylation is often considered an initial event in AD pathogenesis, research in CTE indicates that acetylation at K280 can precede phosphorylation (17). Acetylation at lysine residues such as K164, K174, K180, and K280 is linked to impaired degradation of phosphorylated tau, contributing to its pathological accumulation (15, 57). Conversely, pseudo-acetylation at residues K163, K280, K281, and K369, as well as modifications at the KIGS motif or K321, is associated with decreased tau phosphorylation (42, 48, 58). Specifically, acetylation at KXGS motifs inhibits phosphorylation at S262 and S356 within these sequences (42). These findings emphasize the complex regulatory interplay between acetylation, phosphorylation, and ubiquitination in tau pathology and underscore the necessity for further research to elucidate their roles in neurodegenerative diseases.

EFFECTS OF TAU ACETYLATION

The effect of tau acetylation is site-specific

Acetylation of lysine residues on tau neutralizes their intrinsic positive charge, playing a pivotal role in modulating tau function. The impact of this modification is highly site-specific, with functional consequences varying by the specific residue. Given that acetylation has become recognized as a major PTM influencing tau biology, comprehensive mapping of acetylation sites across various biological conditions is crucial for

understanding the functional significance of each modification. Here, we examine the acetylation of tau noted in AD and how residue-specific acetylation affects tau function and contributes to AD pathogenesis.

The effect of acetylation on tau protein degradation and turnover

The identification of tau protein as a key component of NFTs in AD and other tauopathies has catalyzed extensive research aimed at understanding the mechanisms responsible for its pathological accumulation and aggregation. Tau homeostasis is maintained by a balance between its synthesis and degradation. Critical for the degradation and recycling of macromolecules, including tau, are the ubiquitin-proteasome system (UPS) and the autophagic-lysosomal pathway (ALP). Under physiological conditions, these pathways effectively clear misfolded tau; however, in the context of tauopathies, tau aggregates exhibit resistance to degradation, which leads to their pathological accumulation. PTMs of tau, particularly acetylation, are implicated in promoting this accumulation by interfering with its normal clearance mechanisms. Acetylation at specific lysine residues—including K163, K174, K180, K274, K280, and K281—has been shown to impair tau degradation (15, 59). Min *et al.* proposed a model wherein SIRT1, through its deacetylase activity, removes acetyl groups from tau, thereby exposing lysine residues for ubiquitination. This modification promotes tau degradation via the UPS, thus enhancing tau turnover and facilitating the clearance of phosphorylated tau species (15). Supporting this hypothesis, subsequent research demonstrated that the expression of acetylation-mimicking mutants, such as K174Q in human tau (hTau), reduced tau turnover by inhibiting polyubiquitination in mouse models (16). Additionally, the expression of the K280Q acetylation mimic in transgenic *Drosophila* led to impaired tau clearance (57). Moreover, in the context of autophagic-lysosomal degradation, acetylation at K174, K274, K280, and K281 has been found to promote tau accumulation by redirecting its degradation away from chaperone-mediated autophagy (CMA) towards alternative lysosomal pathways, particularly macroautophagy and endosomal microautophagy (59). This shift is partially attributed to acetylation-induced alterations in tau's pH-dependent binding affinity for the Hsc70 chaperone, which impedes its translocation into the lysosomal lumen while maintaining its membrane association. Consequently, CMA-mediated tau clearance is compromised, leading to intracellular tau accumulation and a significant reduction in CMA efficiency (59).

Conversely, acetylation at specific sites may facilitate tau clearance by preventing the formation of degradation-resistant tau species. Specifically, acetylation within the KXGS motif at residues K259, K290, K321, and K353 interrupts phosphorylation at serine residues S262 and S356 (42). Although these modifications do not target the same residue, they function in a mutually exclusive manner. Phosphorylation at the 12E8

epitope (S262/S356), mediated by kinases such as Par-1/MARK2, diminishes tau's susceptibility to ubiquitin tagging and subsequent proteasomal degradation. This resistance stems from impaired recognition and binding by the ubiquitin ligase CHIP (60). Tau species phosphorylated at these sites also exhibit resistance to degradation, even when chaperone activation via Hsp90 inhibition is induced (60, 61). HDAC6 has been shown to enhance Par-1/MARK2-mediated phosphorylation at KXGS motifs, an effect that is mitigated when these motifs are pseudoacetylated (42). Notably, tau isolated from AD patients and from the tauopathy rTg4510 mouse models shows decreased acetylation and increased phosphorylation at KXGS motifs, a pattern that correlates with reduced tau clearance and increased disease progression (62). Further supporting this link, tau purified from post-mortem AD brain tissue exhibits ubiquitination at KXGS motifs, a modification that likely inhibits acetylation at these lysine residues (63). These findings highlight the critical role of the acetylation-phosphorylation balance at KXGS motifs in regulating tau homeostasis and its implications for AD.

The effect of acetylation on tau solubility and aggregation

Aberrant tau aggregation is a hallmark of AD and various other tauopathies. A growing body of evidence suggests that acetylation significantly influences tau's propensity to aggregate, generally promoting its pathological characteristics. Nonetheless, the precise relationship between tau acetylation and aggregation remains poorly understood, particularly due to site-specific variability among lysine residues. Recent studies suggest that acetylation may precede aggregation, potentially acting as an upstream regulatory mechanism. However, whether acetylation facilitates, impedes, or has a neutral effect on tau aggregation is still being actively researched. Notably, acetylation within the MTBR region has been shown to induce conformational changes that can either enhance or suppress tau aggregation, highlighting the complex and context-dependent nature of this PTM (64).

Acetylation of tau at specific lysine residues has been demonstrated to enhance its aggregation propensity and promote template-dependent misfolding (65). Tau fibrilization is primarily driven by short amyloidogenic motifs (²⁷⁵VQIINK²⁸⁰, ³⁰⁶VQIVYK³¹¹, and ³³⁷VEVKSE³⁴²), which engage in heterotypic interactions with other tau regions to facilitate aggregate formation. Acetylation within the MTBR, particularly near key aggregation-prone sequences (²⁷⁵VQIINK²⁸⁰ and ³⁰⁶VQIVYK³¹¹), is associated with increased formation of pathological tau aggregates (28). Notably, acetylation at K280 correlates with the presence of insoluble tau species in the brains of AD patients and transgenic PS19 and PS19/PDAPP mouse models (14, 38), implicating this modification in aggregation promotion (46). Furthermore, p300-mediated acetylation of tau at multiple sites promotes phosphorylation at S202/T205, T231, and S422 in a cellular model. This effect is abrogated by mutating K280 to alanine (K280A), indicating a critical role for K280 acetylation

in generating aggregation-prone phosphorylated tau species (66). In contrast, another study using an *in vitro* cellular model demonstrated that concurrent acetylation at K280 and K281 reduces phosphorylation at S202/T205, potentially due to increased dissociation of tau from microtubules. This dissociation may enhance access to protein phosphatase 2A (PP2A), facilitating dephosphorylation (65). Despite the reduction in phosphorylated tau species, acetylation mimics at K280/K281 remained capable of forming fibrils displaying ultrastructural properties similar to those observed in the aggregation-prone P301L tau mutant. These mimics promoted β -sheet transitions, thereby enhancing fibrillar aggregation (65). Furthermore, in a transgenic *Drosophila* model expressing the K280Q mutant hTau, phosphorylation was altered in a site-specific manner, though the overall solubility of tau remained unchanged (57).

Additional acetylation sites beyond K280 have been implicated in tau aggregation; specifically, acetylation at K174 promotes tau fibrillization (16). Conversely, acetylation at K274 is associated with the formation of neuritic plaques in the inferior temporal cortex in patients with AD and other tauopathies (39).

Conversely, tau acetylation has been linked to decreased aggregation in certain experimental settings. CBP-catalyzed acetylation at multiple sites has demonstrated the ability to inhibit fibril formation in recombinant tau proteins (44). Similarly, p300-catalyzed acetylation at multiple sites, including K280 and K311, hinders tau aggregation by disrupting liquid-liquid phase separation *in vitro*, a mechanism thought to promote fibrillization under pathological conditions (43). These observations align with the proposed role of HDAC6-regulated acetylation at residues within the KXGS motif—specifically K259, K290, K321, and K353—which inhibits tau phosphorylation and thus limits filament formation and aggregation (42). Unlike lysine residues whose acetylation occurs exclusively in tauopathy brains (16, 38, 39, 67), the four KXGS-associated sites are typically acetylated under physiological conditions, yet demonstrate decreased acetylation in disease states (42). Notably, acetylation-mimic mutations at K321, an HDAC6-responsive site, have been shown to decrease fibril formation in recombinant tau, and further acetylation-mimic mutations at K321 and K353 (K321Q and K353Q), whether alone or combined, are shown to diminish intrinsic tau aggregation and alter the conformational properties of P301L/S320F tau aggregates *in vitro* (68). These findings suggest a protective role for acetylation at these residues, not only by limiting hyperphosphorylation in the disease's early stages but also by impeding the aggregation of pathogenic tau species in later stages.

Moreover, acetylation has been identified as a critical regulatory mechanism that influences tau aggregation in an isoform-dependent manner (69). Specifically, acetylation inhibits the aggregation of 4R tau, while concurrently facilitating amyloid formation in 3R tau (69). Among the acetylated residues, K298 has been identified as a critical determinant in directing this selective aggregation, indicating that site-specific acetylation can alter tau conformation and influence its aggregation po-

tential depending on the isoform (69). This observation may provide a mechanistic basis for the differential accumulation of 3R tau observed in the brains of patients with Pick's disease (69).

It is important to recognize that discrepancies in the reported effects of acetylation at a specific lysine residue often arise from contextual variables, including the experimental model employed, the method used to introduce the modification, and the complex interplay of co-existing post-translational modifications.

The effect of acetylation on tau propagation

Tau protein seeding and propagation are pivotal components of tauopathy. Seeding involves a pathogenic templating mechanism wherein misfolded tau induces conformational changes in native tau, thereby promoting misfolding and aggregation (70). In tauopathies, including AD, misfolded tau aggregates initially form in the entorhinal cortex and subsequently spread to the hippocampus, in accordance with the progression stages outlined by Braak staging (6, 71). Tau propagation encompasses a series of steps including the release of tau from donor neurons, potential extracellular aggregation (either before or after secretion), uptake by recipient neurons, and the induction of tau aggregation within these cells. Despite significant advances made in studying tau propagation through experimental models, the precise molecular mechanisms underlying this process in the human brain have yet to be fully elucidated (6).

The influence of tau acetylation on its propagation appears to vary depending on the specific lysine residue modified. While some studies indicate that acetylation facilitates tau propagation, others demonstrate inhibitory effects. *In vivo* studies have shown that acetylation at residues K174, K274, K280, and K281 promotes the temporal spreading of tau from the entorhinal cortex to the hippocampus, contributing to progressive cognitive deficits (72). Similarly, overexpression of SIRT1, which specifically deacetylates tau at K174 in PS19 transgenic mice, has been observed to reduce K174-acetylated tau levels and accordingly attenuate fibril-induced tau propagation (47).

Mechanistically, tau acetylation affects its extracellular release and propagation by altering degradation pathways. As previously mentioned, acetylation at K174, K274, K280, and K281 hinders tau degradation via CMA, redirecting it to alternate lysosomal pathways such as endosomal microautophagy (59). While acetylation interferes with tau translocation into the lysosome lumen, it promotes trafficking to late endosomes and multivesicular bodies (LE/MVBs), where degradation occurs via endosomal microautophagy. This alteration is linked with an increase in extracellular tau release, facilitating its intercellular spread. Supporting these observations, CMA-deficient mouse models display increased tau transmission, highlighting the significant role of acetylation in tau trafficking and dissemination (59). In line with these observations, hyperactivation of the p300/CBP enzyme impairs the autophagy-lysosomal flux, resulting in increased secretion of both total tau and AT8-positive p-tau, thereby accelerating tau propagation in

primary neurons and PS19 mice (50). Tau acetylation not only enhances its release but may also promote pathology by sensitizing recipient neurons to extracellular seed-induced tau aggregation. Notably, *in vitro* studies demonstrate that acetylation at K280/K281 increases the propensity for the formation of aggregation-prone tau species after fibril seeding (65).

On the other hand, certain acetylation modifications may inhibit tau propagation. The propagation of pathogenic tau can be halted within the recipient neuron by limiting either the internalization of extracellular seeds or the templated aggregation that ensues following their entry. Emerging evidence suggests that lysine acetylation modulates both mechanisms. The uptake of tau via the low-density lipoprotein receptor-related protein 1 (LRP1) relies on positively charged lysine residues within the MTBR (73); chemical capping of these residues or replacing them with acetyl-mimetic glutamine significantly reduces LRP1-mediated internalization (73, 74). Correspondingly, tau constructs carrying AD-linked PTM patterns that include multisite phosphorylation and acetyl-mimetic substitutions at K281, K343, and K353 show decreased affinity for LRP1 and reduced cellular uptake of tau (75). Acetylation also affects the subsequent seed-induced aggregation. Acetylation-mimetic substitutions at five lysine residues within the filament core (K311, K353, K369, K370, and K375), often observed in AD patients, have been demonstrated to inhibit tau aggregation in response to seeds from AD and PSP brains, indicating a potential protective role against seeded propagation (76). However, acetylation-nullifying substitutions (K→R or K→A) do not seem to affect the ability of tau to seed, but rather promote spontaneous self-aggregation resulting in amorphous filament deposits (76). Additionally, each single acetylation-mimetic mutation K259Q, K290Q, K321Q, and K353Q has been found to resist seed-induced tau aggregation *in vitro*. Notably, K321Q and K353Q, whether implemented individually or together, effectively inhibit seed-dependent aggregation, even in the presence of the pathogenic P301L tau mutation (68). Collectively, the current data suggest that tau acetylation can either promote or inhibit prion-like propagation, according to the specific lysine residue altered and its effects on degradation, uptake, and seeding pathways. Discrepancies across studies likely arise from the complex PTM landscape of pathological tau, with the overall effect reflecting the cumulative impact of all PTMs at a particular disease stage. Moreover, tau propagation involves multiple steps—release, uptake, seeding, and intracellular trafficking—thus, an acetylation event that impedes one step (e.g., LRP1-mediated uptake) might be counterbalanced by another modification that enhances a subsequent stage (e.g., seed-induced aggregation). Therefore, evaluating the pathogenic impact of any single PTM necessitates an understanding of its role within the propagation cascade and the collective PTM context in which it occurs.

The effect of acetylation on tau microtubule-binding affinity

The primary physiological function of tau is to stabilize mic-

rotubules and facilitate their assembly through direct interactions mediated by its MTBR. This interaction is crucial for maintaining the structural integrity and transport mechanisms within neurons. In AD, acetylation frequently occurs in the 4R region of the MTBR (18). Acetylation of lysine residues within the MTBR neutralizes their positive charges, consequently impairing tau's affinity for microtubules. This alteration may lead to a loss-of-function mechanism that contributes to neuronal dysfunction in tauopathies (43, 48, 58, 65).

Acetylation-mimetic mutants such as K163/280/281/369Q and K280/K281Q have been shown to decrease tau-mediated microtubule assembly and promote microtubule destabilization *in vitro* (65). Moreover, a *Drosophila* model expressing the 4KQ (K163/280/281/369Q) acetyl-mimic variant demonstrates reduced microtubule binding, even in the context of globally reduced tau phosphorylation, a condition known to independently impair microtubule association (58). These findings underscore the direct and potentially pathogenic impact of tau acetylation on its microtubule-binding capabilities.

Dissociation of tau from microtubules is proposed to expand the pool of cytosolic tau, thereby initiating its pathological accumulation and aggregation. Indeed, certain acetylated residues that enhance fibrillization support this hypothesis (38, 65). However, the relationship between tau-microtubule binding and aggregation is highly site-dependent. For example, acetylation at K174 induces tau aggregation but does not significantly change its affinity for microtubules (16). In contrast, acetylation at residues like K259, K290, K321, and K353 simultaneously decreases microtubule binding and reduces tau aggregation in cell-based assays (68).

Intriguingly, acetylation at K321 has been shown to prevent phosphorylation at S324, a pathological event observed in AD and rTg4510 tauopathy mouse models, known to compromise tau's ability to stabilize microtubules. This acetylation-mediated inhibition could help preserve tau's microtubule-stabilizing function by preventing detrimental phosphorylation events (48). These findings underscore the complex, site-specific effects of tau acetylation on its structural and functional properties.

The effect of tau acetylation on tau synaptic plasticity and AD-related memory deficits

Hippocampal neurodegeneration and subsequent memory loss are prominent clinical features of AD. Consistent with this, spatial memory impairments observed in tauopathy mouse models have been attributed to hippocampal synaptic dysfunction, with accumulating evidence identifying tau acetylation as a key contributor to this deficit (15, 16, 46, 77).

Under physiological conditions, tau exhibits a highly polarized neuronal distribution, predominantly localized to axons (1). However, its redistribution to somatodendritic compartments represents an early pathological event in AD and other neurodegenerative diseases (78). Several factors—including exposure to A β —have been shown to promote this aberrant mislocalization (79–81), with tau acetylation emerging as one

of the driving mechanisms (46, 77, 82).

A double acetylation mimic at lysines 274 and 281 induces tau mislocalization to dendrites in both transgenic mice and primary neurons by destabilizing the axon initial segment (AIS), thereby compromising the barrier that normally restricts tau to axons (82). Acetylation at K280 leads to tau accumulation within neuritic tau beads—pathological aggregates enriched in aged or neuroinflammatory conditions. Inhibition of HDAC6 reduces bead formation, underscoring a pathogenic role for tau acetylation at this site (46).

Mislocalized acetylated tau disrupts postsynaptic signaling. Mice expressing the K274/281Q tau variant display impaired hippocampal long term potentiation (LTP) and memory deficits. Mechanistically, K274/281Q tau reduces the levels of the postsynaptic Kidney/BRAIn (KIBRA) protein, a key regulator of memory function, impairing activity-induced postsynaptic actin remodeling and AMPA receptor insertion (77). This pathogenic mechanism is distinct from the phosphorylation-driven toxicity, which primarily results in the downregulation of glutamate receptors (79). Additionally, acetylation at K174 has been related to the postsynaptic mislocalization of tau. Studies in PS19 mice show that brain-specific deletion of SIRT1 increases K174 acetylation, reduces synaptophysin levels in the cor-

tex and hippocampus, and exacerbates behavioral deficits (47).

Tau acetylation has also been implicated in the accelerating structural neurodegeneration. The K174Q acetylation mimic exacerbates hippocampal atrophy *in vivo* (16), while K280Q enhances photoreceptor degeneration and induces locomotor defects in *Drosophila* models (57). *In vitro*, acetylation mimics at K274 and K281 suppress mitochondrial biogenesis, down-regulate key mitochondrial fusion proteins, and exacerbate mitochondrial dysfunction. *In vivo*, these acetylation mimetics intensify neuronal loss and cognitive decline (83). Further evidence from models of A β -induced tau pathology supports a pathological role for tau acetylation in neurodegenerative processes. Specifically, exposure to A β ₁₋₄₂ increases nitric oxide production, leading to S-nitrosylation of glyceraldehyde-3-phosphate dehydrogenase (GAPDH). S-nitrosylated GAPDH activates p300 and inhibits SIRT1, elevating tau acetylation at K280, which correlates with memory deficits and motor impairment. Pharmacological inhibition of GAPDH S-nitrosylation reduces K280 acetylation and ameliorates associated behavioral impairments, indicating an upstream mechanism by which A β drives tau toxicity (41).

However, tau acetylation may exert protective effects under certain conditions. In the *Drosophila* 4KQ model, acetylated

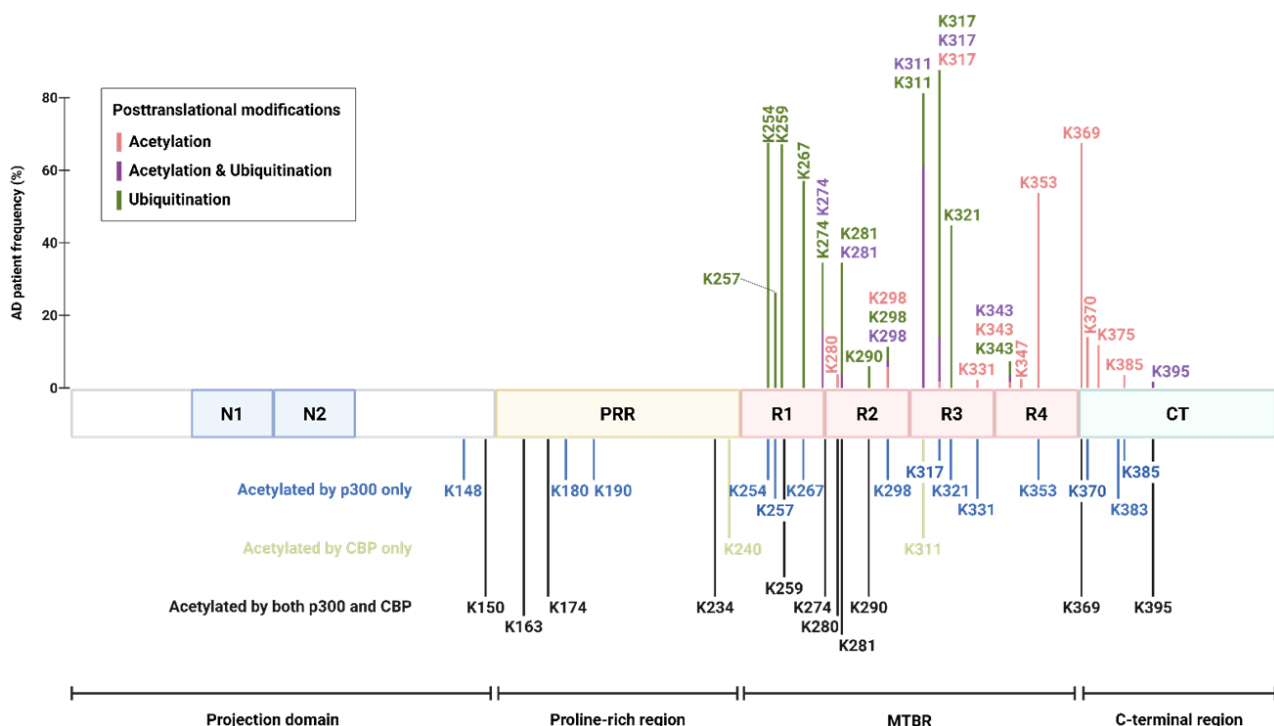


Fig. 1. Structure of tau 2N4R and its acetylation sites in AD. PTMs on lysine residues of tau are mapped to the 2N4R isoform (UniProtKB P10636-8). In the upper panel, line length reflects the relative frequency of each PTM in AD brains. Acetylation (pink) and ubiquitination (green) predominantly occur within the MTBR, often at overlapping sites (purple). Data are adapted from (18). The lower panel depicts lysine residues selectively acetylated by p300 (blue), CBP (light green), or by both enzymes (black) (15, 38, 97).

tau does not exacerbate neurodegeneration but instead reduces levels of phosphorylated tau at the AT8 epitope, improves motor performance, and mitigates A β toxicity, suggesting that specific patterns of tau acetylation can confer neuroprotection (58).

Tau acetylation induces site- and context-specific effects on tau distribution, synaptic function, and neuronal viability. While certain acetylation events compromise tau's physiological roles and facilitate neurodegeneration, others may mitigate pathological processes or fulfill compensatory functions.

THERAPIES TARGETING TAU ACETYLATION

Acetylated tau as a therapeutic target for AD

Although tau acetylation manifests both protective and harmful effects in the context of AD and other tauopathies, its influence remains highly site-specific. Since acetylation of certain lysine residues is pharmacologically modifiable and has the potential to reduce tau pathology, selectively targeting these residues offers a promising therapeutic approach. This strategy involves modulating the activity of enzymes that regulate site-specific acetylation or deacetylation, or using site-specific antibodies to target acetylated tau species. Several such approaches have advanced to clinical trials, underscoring their potential in developing effective therapies for AD.

Current therapies targeting tau acetylation in clinical trials

Salsalate: Salsalate, a small-molecule non-steroidal anti-inflammatory drug (NSAID), has been shown to inhibit tau acetylation by targeting the p300 acetyltransferase pathway (16, 84). Epidemiological studies have suggested its potential neuroprotective effects by associating reduced incidence of AD with salsalate use (85-87). In PS19 transgenic mice, post-symptom-onset administration of salsalate repressed p300 activity, decreased levels of total and K174-acetylated tau, prevented hippocampal atrophy, and ameliorated memory deficits (16). Similarly, in a traumatic brain injury (TBI) mouse model, low-dose salsalate treatment curtailed p300/CBP acetyltransferase activity, thereby reducing tau acetylation, neurodegeneration, and behavioral impairments (84). Clinical data further indicate that individuals treated with p300/CBP inhibitors, such as salsalate or diflunisal, have a lower incidence of both AD and clinically diagnosed TBI (84). In a phase I open-label clinical trial (NCT02422485) involving patients with PSP, salsalate was well tolerated but failed to show significant improvement in clinical symptoms or biomarker profiles (88). A subsequent phase Ib trial (NCT03277573), completed in December 2021, assessed the safety, tolerability, adverse effects, cerebrospinal fluid (CSF) biomarkers, neuroimaging findings, and cognitive outcomes in patients with mild to moderate AD (89). Although salsalate demonstrated a favorable safety profile, it failed to show significant improvements in CSF biomarkers or clinical measures. Additionally, tau positron emission tomography (PET) imaging was not performed to confirm target engagement. The researchers also pointed out that high dropout

rates and baseline variances between the study groups might have skewed the results (89, 90).

Adel Y-01: The monoclonal antibody Y01 selectively targets tau acetylated at K280 and has demonstrated significant therapeutic benefits in transgenic mouse models, which include reductions in tau accumulation and propagation, as well as enhancements in cognitive performance (91). These promising preclinical results have led to the evaluation of Y01 in a Phase I clinical trial (NCT06247345) aimed at assessing its safety and tolerability in healthy participants and those with mild cognitive impairment (MCI) and mild AD. The outcomes of these ongoing clinical trials are still pending. Significantly, Y01 is the first antibody designed specifically to target a defined site of tau acetylation, representing a novel strategy for disease modification. The clinical trials of Y01 may yield important insights into the viability and therapeutic potential of targeting site-specific lysine acetylation in tauopathies.

Current therapies under development targeting tau acetylation

Diflunisal: Diflunisal has generated interest due to its potent inhibitory activity against the acetyltransferases p300 and CBP, demonstrating greater efficacy than salsalate (92). Although its ability to cross the blood-brain barrier (BBB) is relatively limited, diflunisal suppresses p300/CBP activity at significantly lower concentrations, enhancing its therapeutic relevance in modulating tau acetylation (92). Large-scale, population-based analyses provide solid epidemiological evidence supporting the neuroprotective potential of diflunisal. In a cohort of over 7.23 million commercially insured individuals in the United States, long-term use of diflunisal was associated with a substantial reduction in the incidence of both AD and clinically diagnosed TBI (84). Specifically, the hazard ratio for AD among diflunisal users was 0.17 over a six-year follow-up period, indicating a markedly reduced risk compared to non-users and surpassing the protective effects observed with salsalate (84). Complementary preclinical data further bolster its therapeutic potential. In murine models, diflunisal was shown to cross the BBB and significantly reduce levels of acetylated tau in cortical brain tissue (84). These reductions in brain acetylated tau levels coincided with corresponding decreases in plasma acetylated tau, suggesting effective modulation of pathological tau acetylation at both central and peripheral levels (84). Collectively, these findings highlight the promise of diflunisal as a pharmacological agent for tauopathies. Although it is not currently under investigation in clinical trials for AD, the convergence of mechanistic, preclinical, and epidemiological evidence strongly supports further exploration of diflunisal as a disease-modifying therapy targeting tau acetylation.

Acetylated tau K174 antibody: A monoclonal antibody selectively targeting tau acetylated at lysine 174 (ac-tauK174) has demonstrated disease-modifying effects in preclinical models (93). In PS19 transgenic mice, treatment with anti-ac-tauK174 led to a reduction in pathological tau burden, preservation of

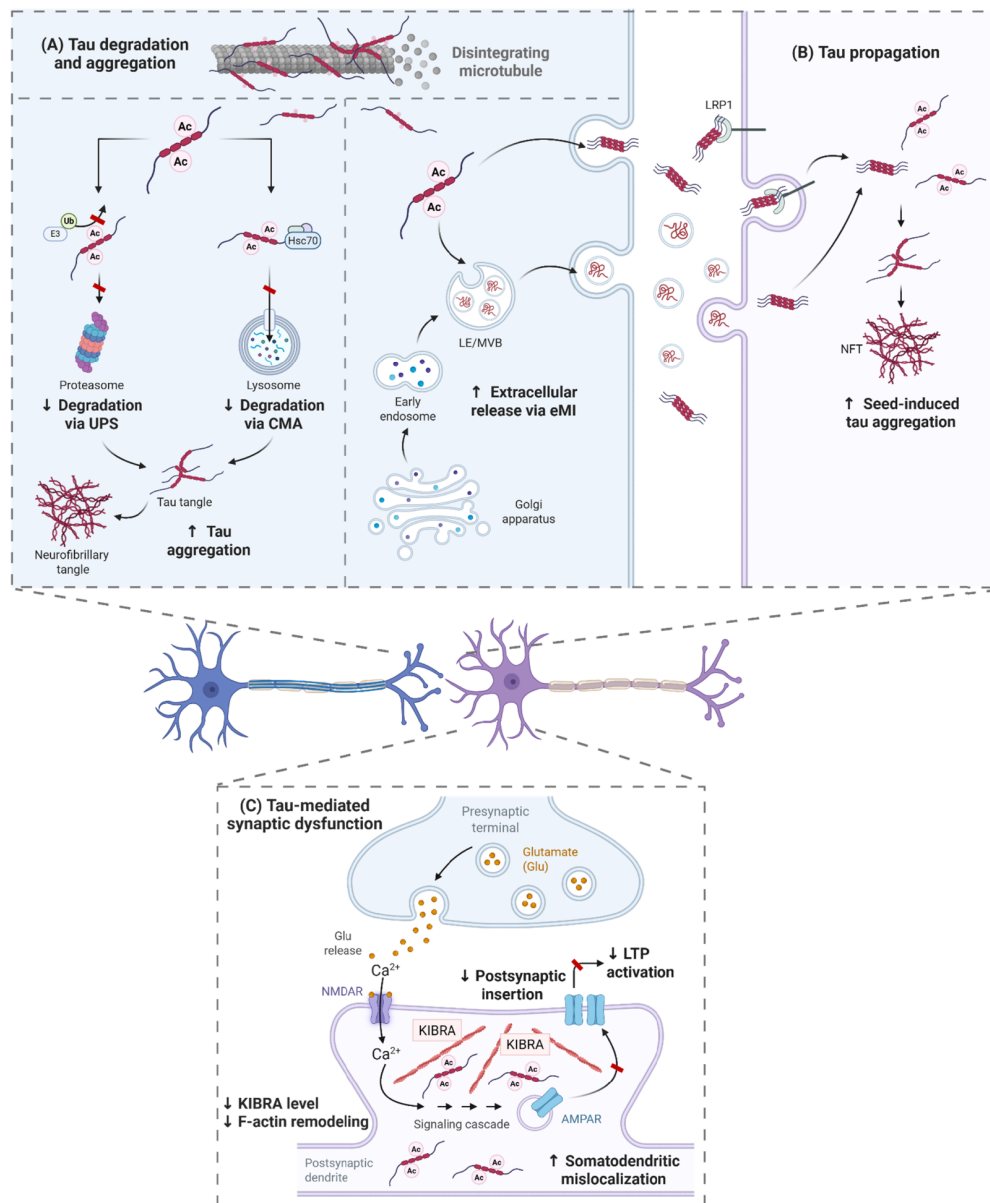


Fig. 2. The pathological roles of tau acetylation in AD. An illustration of acetylation-induced functional alterations of tau that exacerbate AD pathology through impaired degradation, increased aggregation and propagation, and synaptic dysfunction. (A) Acetylation impairs tau degradation and promotes aggregation. Acetylation at lysine residues of tau obstructs its ubiquitination, thereby preventing degradation via the UPS. Furthermore, acetylation hinders tau's entry into lysosomes, consequently impeding its degradation through CMA. This reduction in tau degradation leads to an accumulation of cytosolic tau, increasing its propensity for pathogenic aggregation. (B) Acetylation accelerates tau propagation and enhances seed-induced aggregation. When CMA-mediated tau degradation is compromised due to acetylation, tau is redirected towards alternative lysosomal pathways, such as e-MI, which degrades proteins within LE/MVBs. Increased translocation of tau into LE/MVBs facilitates its extracellular release, thereby promoting intercellular spread. Once released into the extracellular space, acetylated tau can enter recipient cells through various mechanisms, including receptor-mediated endocytosis and direct endocytosis. Recipient cells containing acetylated tau become more susceptible to seed-induced aggregation, significantly contributing to the dissemination of tauopathy throughout the brain. (C) Acetylated tau induces synaptic dysfunction. Acetylated tau undergoes pathological relocation from the axonal compartment to somatodendritic regions. This mislocalized tau reduces the levels of the postsynaptic KIBRA protein, which in turn impairs activity-induced postsynaptic F-actin remodeling and AMPA receptor insertion. Ultimately, this cascade disrupts LTP, a critical mechanism for synaptic plasticity, leading to observable memory deficits.

Table 1. Site-specific effects of acetylation on tau function in AD

Site of acetylation	Domain of tau	Effect on tau function	Experimental models	Ref.
K311/353/369/370/375	MTBR C-terminal	Suppresses tau aggregation when exposed to seeds derived from AD and PSP Decreases MT binding affinity Acetylation-nullifying substitutions do not affect tau seeding ability but promote spontaneous self-aggregation	HEK293T cells	(76)
K163/280/281/369	PRR MTBR	Found in CBP mediated acetylation Impairs MT assembly Promotes tau fibrillization	<i>In vitro</i> HEK-T40 cells QBI-293 cells WT, PS19, PS19/PDAPP mouse AD, CBP, PiD brain	(38)
		Specifically increases p-S262 while decreasing tau1, AT8, AT100, and PHF-1 p-tau Hinders binding to microtubules No detrimental neurodegeneration Ameliorates A β toxicity; decreases AT8 positive p-tau levels, improves neuronal and locomotor	hTau KI <i>Drosophila</i>	(58)
K174/274/280/281	PRR MTBR	Redirects tau degradation from CMA to macroautophagy and endosomal microautophagy Elevates tau accumulation Increases extracellular tau release Accelerates intercellular propagation Promotes tau propagation Leads to tau accumulation, glial activation, synaptic dysfunction, and cognitive impairments	<i>In vitro</i> Neuro-2a (mouse neuroblastoma cell line) iPSC-derived human neurons WT, tg mouse AD brain Tau induced mouse	(59) (72)
K259/290/321/353	MTBR	KXGS motif Inhibits tau filament assembly, aggregation, and accumulation Disrupts phosphorylation at S262 and S356 which form p-tau species that are resistant to degradation	<i>In vitro</i> HEK-293T cells HeLa cells WT, rTg4510 mouse AD brain	(42)
K163/174/180	PRR	Hinders tau degradation by disrupting ubiquitination Increases accumulation of AT8 p-tau	<i>In vitro</i> HEK293T cells Rat primary neurons WT, tg, PS19 mouse AD brain	(15)
K281/343/353	MTBR	Included in tau species with AD-linked PTM patterns (phosphorylation and acetylation mimetics) Does not affect the level of tau released from cells Reduces uptake into cells Reduces binding to LRP1	CHO cells iPSC-derived neurons	(75)
K274/281	MTBR	Triggers LTP deficits and impairment of memory formation Induces loss of postsynaptic protein KIBRA Reduces AMPA receptor trafficking Induces tau somatodendritic mislocalization	Rat primary neurons WT, tg, hAPP-J20, rTg4510 mouse AD brain HeLa cells Rat primary neurons WT, tg mouse AD brain	(77) (82)
		Exacerbates cognitive impairments with neuronal loss, reduced dendritic spines and neuronal dendritic complexity Exacerbates mitochondrial dysfunctions Reduces mitochondrial biogenesis Disrupts mitochondrial dynamics by reducing mitochondrial fusion	Rat hippocampal primary neurons Tau induced mouse	(83)
K280/K281	MTBR	Impairs tau-mediated MT assembly Reduces phosphorylation at S202/T205 Enhances tau fibrillar aggregation by accelerating β -sheet structure formations Enhances seed-dependent tau aggregation and p-S396 at soluble fraction	<i>In vitro</i> QBI-293 cells Mouse primary cortical neurons	(65)
K280/311	MTBR	Located in critical aggregation-prone sequences embedded in the MTBR Associates with increased formation of pathological tau aggregates	<i>In vitro</i> HEK293T cells	(28)

Table 1. Continued

Site of acetylation	Domain of tau	Effect on tau function	Experimental models	Ref.
K321/K353	MTBR	Inhibits intrinsic aggregation Alters conformational amyloid structure of P301L/S320F tau aggregates Suppresses seeded-induced aggregation even with P301L tau mutation present	HEK293T cells	(68)
K174	PRR	Decreases overall tau turnover and leads to tau accumulation by suppressing polyubiquitination Induces neurodegeneration and cognitive deficits	<i>In vitro</i> HEK293T cells Mouse, rat primary neurons WT, K174Q hTau induced, PS19 mouse AD brain	(16)
		Specifically elevated by brain-specific SIRT1 deletion in tauP301S mice Leads to increased mislocalization of tau to postsynaptic dendritic compartment Deficiency of SIRT1 associates with increased mortality, synaptic loss, and behavioral impairments Reduced by overexpression of SIRT1 Reduces tau spread and enhances synaptic function	HEK293T cells WT, tg, tauP301S mouse	(47)
K259	MTBR	Resists seed-induced tau aggregation Decreases MT binding	HEK293T cells	(68)
K280	MTBR	Induced in response to exposure to A β ₁₋₄₂ following NO production Leads to memory impairment, and locomotor dysfunction	HEK293T cells Mouse primary cortical neurons WT, tg mouse AD brain	(41)
		Delays tau clearance Exacerbates photoreceptor neurodegeneration Increases pS262 but not AT8, AT100 p-tau No effect on tau solubility	hTau-K280Q transgenic <i>Drosophila</i>	(57)
		Decreases AT100 p-tau, but not other p-tau species No effect on microtubule binding ability	hTau KI <i>Drosophila</i>	(58)
		Potentially essential for the increase of phosphorylation at S202/T205, T231, S422, but not at S199/202 or S396 induced by p300 mediated tau acetylation	SH-SY5Y (human neuroblastoma cell line)	(66)
		Found in neuritic tau beads induced by A β -induced neuroinflammation	HEK-293A cells Mouse primary neurons WT, tg mouse AD brain	(46)
K290	MTBR	Resists seed-induced tau aggregation Decreases MT binding	HEK293T cells	(68)
K298	MTBR	Key determinant in directing isoform-specific aggregation Inhibits the aggregation of 4R tau Promotes amyloid formation in 3R tau	<i>In vitro</i>	(69)
K321	MTBR	Inhibits recombinant tau fibril formation and aggregation Suppresses phosphorylation at S324, which impairs microtubule assembly capacity	<i>In vitro</i> HEK293T cells Primary mouse neuron WT, P301L htau induced, rTg4510 mouse AD brain	(48)
K321	MTBR	Resists seed-induced tau aggregation Decreases MT binding	HEK293T cells	(68)
K353	MTBR	Resists seed-induced tau aggregation Decreases MT binding	HEK293T cells	(68)

memory function, and protection against TBI-induced neurodegeneration. Furthermore, this antibody reverses TBI-associated transcriptional dysregulation in microglia and oligodendrocytes, indicating its capacity to modulate the neuroinflammatory environment (93). Human data supporting its translational relevance show that plasma levels of ac-tauK174 significantly

increase within 24 hours following TBI, suggesting that this post-translational modification not only serves as a pathological hallmark but also as a potential biomarker of acute tau-mediated neuronal injury (93). These findings reinforce the rationale for site-specific immunotherapeutic strategies aimed at neutralizing toxic tau species before the onset of down-

stream neurodegenerative cascades. The combined therapeutic investigation of diflunisal and the anti-ac-tauK174 antibody epitomizes a dual-modality approach in targeting pathological tau acetylation—through enzymatic inhibition and epitope-specific immunoneutralization. These complementary strategies mark significant progress toward precision medicine in AD and related tauopathies by selectively targeting acetylation-driven tau toxicity.

CONCLUSIONS

Tau acetylation has been recognized as a crucial PTM in AD and related tauopathies, influencing essential aspects of tau biology, such as degradation, aggregation, propagation, microtubule binding, and synaptic function. These effects are highly site-specific as summarized in Fig. 2 and Table 1: acetylation at residues like K174, K274, and K280 promotes tau accumulation and pathology, while modifications at sites such as K321 and K353 may exert protective roles by attenuating phosphorylation and aggregation.

The interplay between acetylation and other PTMs—particularly phosphorylation and ubiquitination—introduces additional complexity into tau regulation. Acetylation can either promote or inhibit the addition of other PTMs, thus affecting tau stability, localization, and toxicity. This intricate crosstalk highlights the necessity of residue-specific characterization to identify therapeutic targets that maximize efficacy while minimizing off-target effects.

Several therapeutic strategies targeting tau acetylation are being explored. Small-molecule inhibitors of the p300/CBP acetyltransferases, such as salsalate and diflunisal, have shown potential in reducing acetylated tau levels and improving tau-related outcomes in preclinical studies, though clinical outcomes have been limited. Meanwhile, monoclonal antibodies targeting acetylated tau (e.g., at K174 or K280) have shown promising effects in animal models of tauopathy and are advancing through early-phase clinical trials.

Despite these advances, many acetylation sites frequently observed in AD brains—such as K317, and K375—remain poorly understood. A more comprehensive understanding of these modifications, including their temporal patterns across disease stages and their interactions with other tau PTMs, is crucial for directing the development of effective therapeutics.

Although recent diagnostic guidelines have incorporated neuroimaging and genetic risk factors, the diagnosis of AD continues to rely primarily on the clinical evaluation of symptoms. This underscores the pressing need for fluid biomarkers that are not only sensitive and scalable but also capable of detecting underlying pathology, staging disease progression, and monitoring therapeutic response. For over four decades, tau has been a central focus in this context. Neuronal tau, upon release into the CSF, leads to consistently elevated levels of total tau and various phosphorylated forms in AD and other neurodegenerative conditions (94). The CSF

A β 42:p-tau181 ratio was approved by the U.S. Food and Drug Administration (FDA) as a diagnostic biomarker for AD in 2022 (95), and more recently, in 2025, the plasma A β 42:p-tau217 ratio received the agency's first clearance for a blood-based *in vitro* diagnostic test for the disease (96). Despite these advances, acetylated tau remains an underexplored biomarker for tauopathies. Given the well-established diagnostic and prognostic utility of phosphorylated tau species (94), a comprehensive investigation of acetylated tau—another post-translational modification implicated in neurodegeneration—is both logical and overdue. Emerging evidence from TBI studies indicates that plasma acetylated tau correlates with neurodegeneration in both human patients and animal models, suggesting that specific acetylation sites may serve as minimally invasive indicators of tau pathology in AD. Characterizing the temporal dynamics and site specificity of tau acetylation across disease stages may, therefore, facilitate the development of a novel class of fluid biomarkers with direct clinical relevance.

In summary, targeting tau acetylation presents a significant therapeutic opportunity in AD and related tauopathies. Future research should focus on elucidating the site-specific functions of acetylation, optimizing targeted drug candidates, and translating preclinical findings into clinically effective interventions.

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CONFLICTS OF INTEREST

The authors have no conflicting interests.

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